

Terry Liu (Liu, Guan-Ting)

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SUMMARY

Quantitative researcher with graduate training in statistics. Ran a live market-making strategy returning **32.1% annualized over four months of real capital**; built deep learning models on tick-level order book data, time-series validation frameworks for CTA research, and Go-based market-data infrastructure running unattended in production.

EXPERIENCE

Quantitative Trading Intern

May 2025 – Dec 2025

Bincentine — Crypto Research & Market Making

Taipei, Taiwan

- Designed and ran a **market-making strategy for low-liquidity assets** on a crypto exchange, using a predictive model to set quote levels and select hedging instruments while dynamically accounting for transaction costs and funding-rate drift. Live Sep–Dec 2025: **32.1% annualized / 8.62% cumulative; 12.1% / 3.25%** excluding outlier events, showing the edge is not tail-dependent.
- Researched cross-market and **DeFi arbitrage**, including options pricing models and funding-rate dislocations across CEX and on-chain venues.
- Implemented dynamic risk management (position sizing, inventory skew, kill-switch) supporting continuous unattended operation.

Quantitative Trading Intern

Apr 2025 – Jul 2025

SQM Technologies — Deep Learning Research

Taipei, Taiwan

- Trained deep learning models on **tick-level Limit Order Book data** for Taiwan equities, focusing on model ensembling and reusable feature-transformation modules that let new LOB-derived feature sets swap into the pipeline without downstream code changes.

Consultant

Mar 2025 – Dec 2025

WorldQuant BRAIN

Remote

- Achieved **Gold Consultant** status through systematic factor analysis and alpha searching across fundamental, price-volume, and analyst datasets; validated signal robustness out-of-sample and cross-region.

SELECTED PROJECTS

Noise-Reduction + ML for Intraday CTA Strategy | *Python, scikit-learn, Time-Series CV* | [GitHub](#)

- Compared three model-update regimes — fixed parameters, rolling window, expanding window — on Taiwan and Hong Kong index futures; expanding-window retraining materially improved out-of-sample stability.
- Applied **label denoising** to suppress high-frequency noise in classification targets, reducing overfitting; strategy outperformed benchmark over a **2000–2024 out-of-sample backtest**.

HFT Order Book Collector | *Go, WebSocket, GCS, systemd, AWS EC2* | [GitHub](#)

- Go service streaming Bybit depth-50 perpetual order book (**20ms updates**) through a 100k-buffered channel, tagging each event with a **nanosecond receive timestamp** and flushing hourly gzip NDJSON partitions to Google Cloud Storage.
- Runs unattended as a systemd unit on EC2 with auto-reconnect, graceful shutdown that flushes partial buffers, cross-compiled release builds, and a write-only scoped service account.

sj-point — Automated Short Straddle System | *Python, PyQt6, Linux, GCP* | [GitHub](#)

- End-to-end automated options system for the Taiwan index options market: data acquisition, statistical strike selection, risk control, signal generation, and order execution as independent modules, deployed on Linux via GCP with secure authentication and a real-time PyQt6 monitoring interface.

EDUCATION

National Yang Ming Chiao Tung University (NYCU)

Sep 2023 – Jun 2025

M.B.A., Institute of Business and Management | Minor: Institute of Statistics | QS World #177

Hsinchu, Taiwan

- Graduation Score **3.97/4.3** | Thesis Defense **4.3/4.3 (full marks)** | Cumulative GPA 3.64/4.3
- Thesis:** Integrating Noise-Reduction Techniques and Machine Learning Models for Enhanced Intraday Trading — Evidence from the Taiwan and Hong Kong Futures Markets
- Statistics coursework:** Statistical Computing, Time Series Analysis, Regression Analysis, Machine Learning, Financial Statistics, Advanced Algorithms, Statistical Methods & Data Analysis, Generative AI

TECHNICAL SKILLS

Languages: Python, Go, SQL, Bash

ML & Statistics: PyTorch, scikit-learn, XGBoost, CNN/LSTM, time-series cross-validation, factor analysis, signal denoising

Trading & Data: LOB microstructure, options pricing, market making, backtesting with cost & slippage, WebSocket/REST market data, tick data pipelines

Infrastructure: Linux, GCP (Compute Engine, GCS), AWS EC2, Docker & Compose, systemd, BoltDB, Git, PyQt6

Certifications: WorldQuant BRAIN Gold Consultant; Qualification Exam for Futures Specialist (Taiwan)